Name:

BUID:

Lab 9: Progress and Preservation

1. Recall the simply-typed lambda calculus (STLC) with small-step CBN semantics.

$$\begin{array}{rcll} e & ::= & \bullet \mid \lambda x^{\tau}.e \mid ee & \text{(expressions)} \\ \tau & ::= & \top \mid \tau \to \tau & \text{(types)} \\ v & ::= & \bullet \mid \lambda x^{\tau}.e & \text{(values)} \end{array}$$

$$\frac{}{\Gamma \vdash \bullet : \top}\ \textsc{unit} \qquad \frac{(x : \tau) \in \Gamma}{\Gamma \vdash x : \tau}\ \textsc{var} \qquad \frac{\Gamma, x : \tau_1 \vdash e : \tau_2}{\Gamma \vdash \lambda x^{\tau_1}.e : \tau_1 \to \tau_2}\ \textsc{fun} \qquad \frac{\Gamma \vdash e_1 : \tau_2 \to \tau \qquad \Gamma \vdash e_2 : \tau_2}{\Gamma \vdash e_1 e_2 : \tau}\ \textsc{app}$$

$$\frac{e_1 \longrightarrow e_1'}{e_1 e_2 \longrightarrow e_1' e_2}\ \textsc{appLeft} \qquad \frac{}{(\lambda x^{\tau}.e_1)e_2 \longrightarrow [e_2/x]e_1}\ \textsc{beta}$$

Suppose we made a mistake when writing the FUN rule, and forgot to require the type annotation to match the argument type:

$$\frac{\Gamma, x : \tau \vdash e : \tau_2}{\Gamma \vdash \lambda x^{\tau}.e : \tau_1 \to \tau_2}\ \textsc{funWrong}$$

Demonstrate that STLC with FUN replaced by FUNWRONG does not satisfy preservation.

2. Suppose we consider a variation of STLC that requires applications to have two arguments.

$$\begin{array}{rcll} e & ::= & \bullet \mid \lambda x^{\tau}.e \mid e(e,e) & \text{(expressions)} \\ \tau & ::= & \top \mid \tau \to \tau & \text{(types)} \\ v & ::= & \bullet \mid \lambda x^{\tau}.e & \text{(values)} \end{array}$$

$$\frac{}{\Gamma \vdash \bullet : \top}\ \textsc{unit} \qquad \frac{(x : \tau) \in \Gamma}{\Gamma \vdash x : \tau}\ \textsc{var} \qquad \frac{\Gamma, x : \tau_1 \vdash e : \tau_2}{\Gamma \vdash \lambda x^{\tau_1}.e : \tau_1 \to \tau_2}\ \textsc{fun} \qquad \frac{\Gamma \vdash e : \tau_1 \to \tau_2 \to \tau \qquad \Gamma \vdash e_1 : \tau_1 \qquad \Gamma \vdash e_2 : \tau_2}{\Gamma \vdash e(e_1, e_2) : \tau}\ \textsc{app}$$

$$\frac{e \longrightarrow e'}{e(e_1, e_2) \longrightarrow e'(e_1, e_2)}\ \textsc{appLeft} \qquad \frac{}{(\lambda x^{\tau_1} \lambda y^{\tau_2}.e)(e_1, e_2) \longrightarrow [e_2/y][e_1/x]e}\ \textsc{beta}$$

Demonstrate that this variant of STLC does not satisfy progress.

3. *Optional Challenge.* Show by induction on derivations that the system from Problem 1 satisfies progress.[1]

4. *Optional Challenge.* Show by induction on derivations that the system from Problem 2 satisfies preservation.[2]

Write your solutions below and on the back on this page.

[1]This challenge problem implies progress does not imply preservation.
[2]This challenge problem implies preservation does not imply progress.

Continue your work here.